Deliverable 6: Master Academic DBMS Project Report

Project: Intelligence Fusion Platform • **Course:** DBMS Mini Project (UE24CS395)
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1. Abstract & Problem Statement

Modern law enforcement and cyber intelligence centers handle two conflicting categories of data: rigid transactional records (custody transfers, suspect identities, court exhibits) requiring ACID consistency, and high-volume, heterogeneous surveillance telemetry (CCTV logs, phone extractions, and encrypted chat dumps) arriving in unstructured formats. Single-database architectures either sacrifice schema rigidity or fail to ingest polymorphic artifacts. The **Intelligence Fusion Platform** resolves this through a verified Polyglot Persistence Architecture, combining PostgreSQL 15+, MongoDB 7.0+, FastAPI, and Next.js.

2. System Architecture & Component Mapping

Tier / Layer	Technology Stack	Operational Responsibility
Frontend UI	Next.js 15+ App Router, TypeScript, Tailwind CSS	Digital forensic workstation, link graph, timeline
REST Gateway	Python FastAPI, Pydantic v2, Uvicorn	Federated cross-database search, JWT auth, RBAC
Relational Store	PostgreSQL 15+ (SQLite Failover)	22 tables, strict FKs, check constraints, BCNF
Document Store	MongoDB 7.0+ (PyMongo Driver)	Polymorphic intelligence records, weighted text search
Tooling & CI	PowerShell, Bash, Docker Compose, Pytest	One-command startup, safe non-destructive seeding

3. Automated Verification & Testing Results

The automated test harness (pytest backend/tests/test_api.py -v) verifies 10 distinct integration endpoints covering relational CRUD, evidence chain of custody, cross-database search, and DBMS concepts with 100% test passage (9 passed, 1 dynamically skipped when MongoDB is offline).